

Upper Respiratory Tract Infection

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Outline

- Upper respiratory tract infections
- Nasopharyngeal carcinoma
- Laryngeal tumors

Common Cold

- It is viral illness in which the symptoms of rhinorrhea and nasal congestion are prominent and systemic symptoms are absent
- Epidemiology: Children have average of 6 to 7 colds/year, 10 to 15% have at least 12/year.
- Decrease with age ,2 to 3/year in adults

Pathogens Associated with Common Cold

Agents primarily associated with colds	Rhinoviruses	Frequent
	Coronaviruses	Occasional
Agents primarily associated with other clinical syndromes that also cause common cold symptoms	Respiratory syncytial virus	Occasional
	Influenza viruses	Uncommon
	Parainfluenza viruses	Uncommon
	Adenoviruses	Uncommon
	Enteroviruses	Uncommon

- RV → peaks early fall(august- October) and late spring (April-May).
- Para influenza viruses peaks - late fall
- RSV and Influenza (December –April)

PATHOGENESIS

Spread :

Viruses spread by small-particle aerosols, large-particle aerosols and direct contact.

- RV and RSV direct contact is more efficient .
- Influenza more spread with the small particle aerosols.
- Influenzas /Adenovirus infection -> destruction of nasal epithelial lining
- Rhinovirus ,Corona viruses and RSV -> no apparent histological damage as in nasal epithelium

Clinical Manifestations

- Onset usually after 1-3 days of acute infection
- Nasal congestion accompanied by watery
- Discharge
- Sneezing
- Scratchy and dry sore throat
- Low grade fever and other constitutional symptoms

Conditions that may mimic the Common Colds

Condition	Differentiating Features
Allergic Rhinitis	Prominent itching and sneezing,nasal eosinophilia
Foreign Body	unilateral ,foul-smelling discharge ,bloody nasal secretions
Sinusitis	Headache,facial pain, periorbital edema,persistence of rhinorrhea or cough > 10-14 days
Streptococcal nasopharyngitis	Nasal discharge that excorites the nares
Pertussis	Onset of persistent or paroxysmal cough
Congenital Syphilis	Persistent rhinorrhea with onset in 1 st 3 mo. Of life

Complications

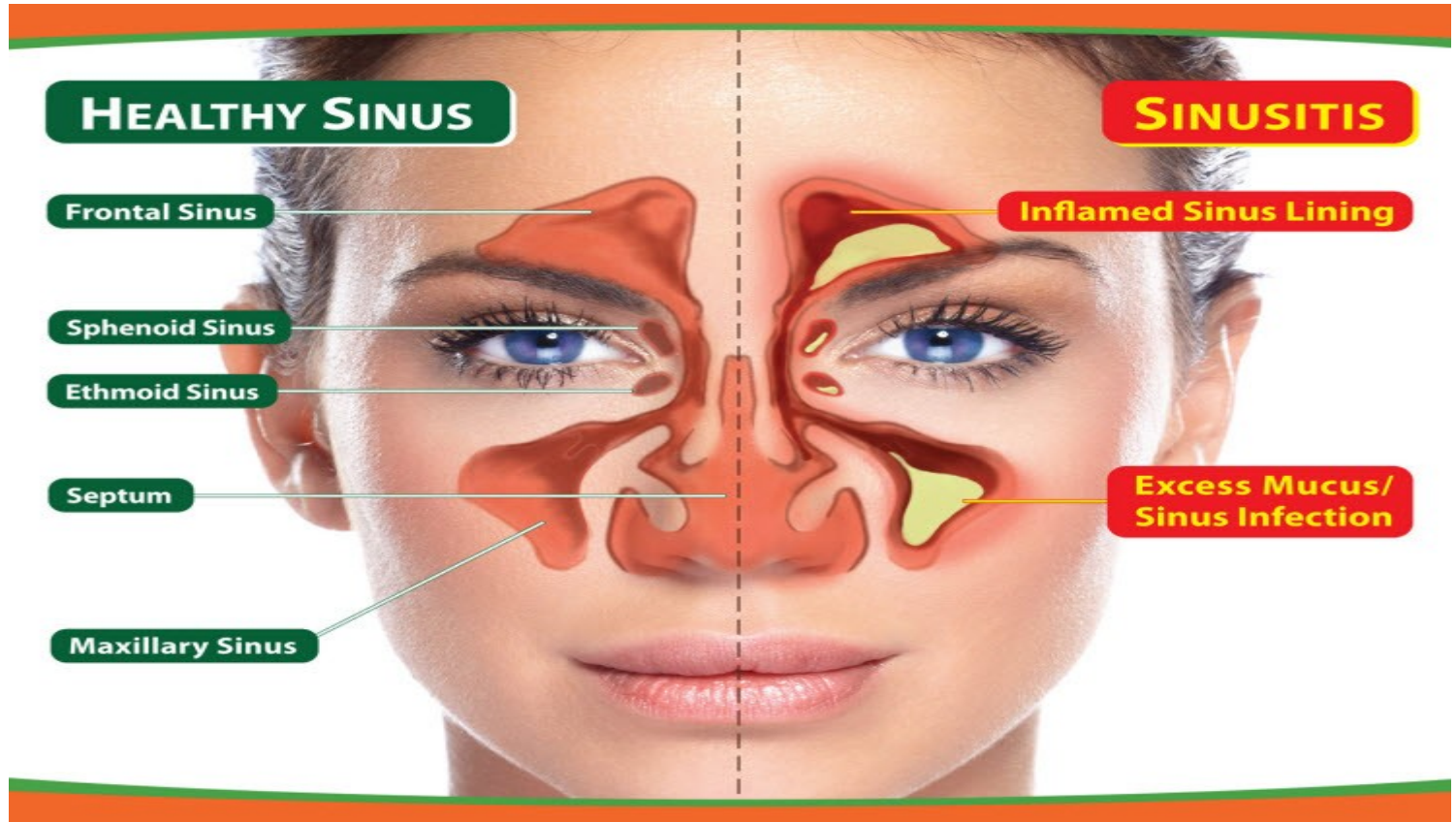
- OTITIS MEDIA
- SINUSITIS
- EXACERBATION of ASTHMA

Sinusitis



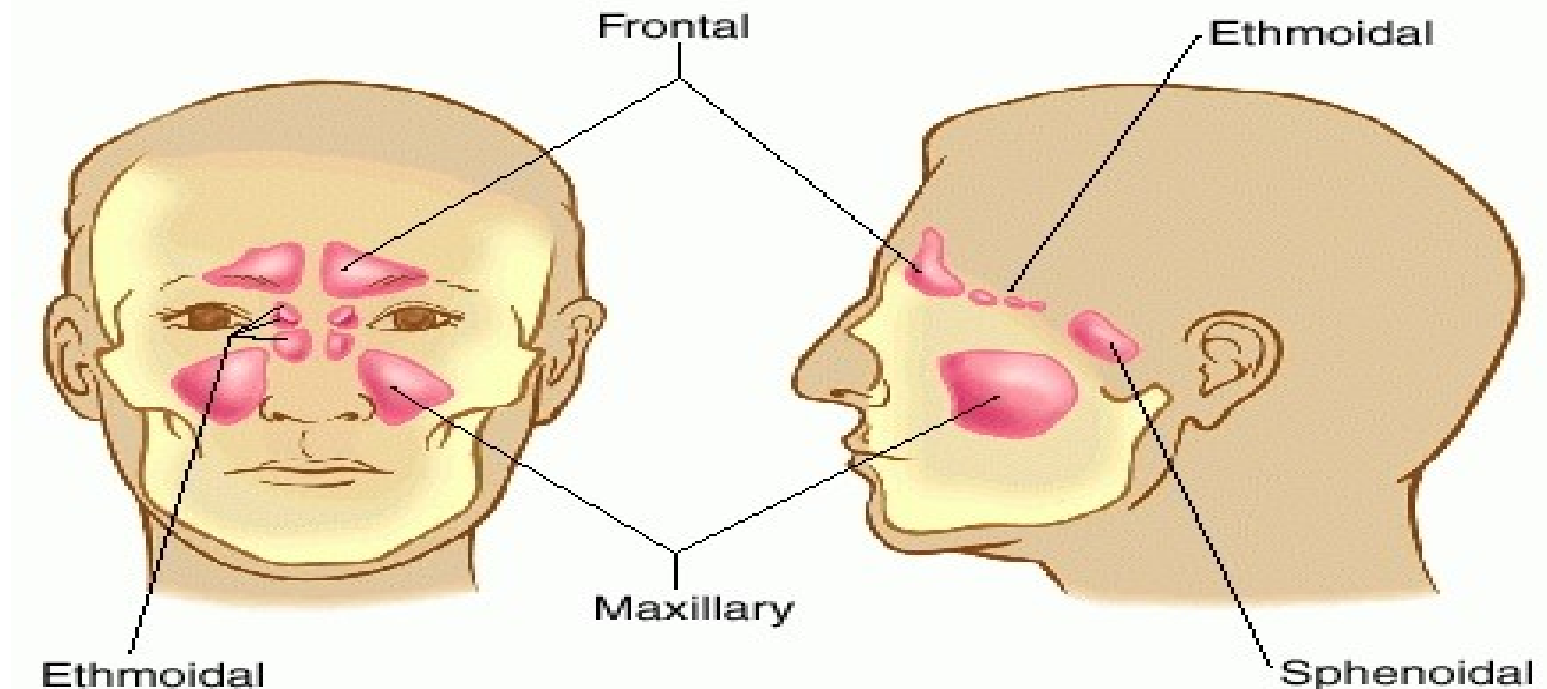
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- Inflammation of the lining of the paranasal sinuses.



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- The maxillary sinuses are the most common site (85%), followed by ethmoidal (65%), sphenoidal (39%), and frontal (32%) involvement



Classification

- Acute sinusitis
< 4 Weeks
- Subacute sinusitis
4 Weeks and < 3 months
- Chronic sinusitis
> 3 Months
- Recurrent acute sinusitis
Diagnosed when 2-4 episodes of infection occur per year.

Etiology

➤ Infection

- Viral (rhinovirus, influenza virus)
- Bacterial (Streptococcus pneumonia)
- Fungal (Rare)

- Cilia in the sinuses do not work properly due to some medical conditions (kartegner syndrome).
- Colds and allergies may cause too much mucus to be made or block the opening of the sinuses.
- A deviated nasal septum, nasal bone spur, or nasal polyps may block the opening of the sinuses

Clinical Presentation

- ✓ Purulent nasal discharge (v. imp) (diagnostic)
- ✓ Nasal airway obstruction
- ✓ Headache, irritability, or facial pain
- ✓ Fever
- ✓ Postnasal drip

Intracranial Complications

1. Meningitis (the most imp)
2. Epidural abscess
3. Subdural abscess
4. Intracerebral abscess
5. Cavernous sinus, venous sinus thrombosis

ACUTE PHARYNGITIS

- Millions of visits to primary care providers each year are for sore throat.
- *The majority of cases of acute pharyngitis are caused by typical respiratory viruses*

Etiology

- A wide variety of organisms cause acute pharyngitis.
- About **30% of cases have no identified cause**
 - ✓ viruses
 - ✓ Group A B-hemolytic strep(GABHS).
 - ✓ Others; group C strept. ,Arcanobacterium hemolyticum,Francisella tularensis,Mycoplasma pneumoniae ,Nissera gonorrhoeae,Corynebacterium diphtheriae .

EPIDEMIOLOGY

- Viral URTI mostly in winter and spring, spread by close contact
- Incidence increases among children then declines late adolescents and adults.
- Group A streptococcal pharyngitis is primarily a disease of children **5–15 years** of age; *it is uncommon among children <3 years old*

Clinical Manifestations

- Onset often rapid; sore throat ,fever.
- P/E :**Erythematous pharynx** ,tonsils enlarge with yellow blood tinged **exudates**, possible to have **petechiae** 'doughnut' lesions on soft palate and post. Pharynx . Uvula-red swollen.
- Ant. Cervical L.N enlarged, tender.
- **Scarlet fever**;circumoral pallor,strawberry tongue ,fine red papular rash 'sand paper'

Viral pharyngitis

- More gradual, more with rhinorrhea ,cough,diarrhea.
- Adenovirus may have concurrent conjunctivitis,fever
- EBV ;prominent tonsillar enlargement,cervical lymphadenitis ,HSM ,fatigue –IM.
- PRIMARY HERPES SIMPLEX ,young children ,high feve ,gingivostomatitis



Complications

- Viral URTI predispose to Middle Ear Infections
- Streptococcal Pharyngitis complications
 - ✓ parapharyngeal abscesses
 - ✓ **AGN** and **Acute Rheumatic Fever**.
- *The risk of rheumatic fever can be reduced by timely penicillin therapy*

ACUTE EPIGLOTTITIS

- It is medical emergency
- Etiology
 - ✓ H.Influenza ,most common before vaccine introduction.(reduced by 90%).
 - ✓ Streptococcus pyogens,S.pneumoniae ,staph. Aureus ,now larger proportion.
- Age was 2-4yr but as early as 1st year and late as 7 years have been seen.

Clinical Manifestations

- ✓ acute fulminating course of high grade fever
- ✓ dyspnea ,sore throat ,
- ✓ rapidly progressive respiratory obstruction
- ✓ difficult swallowing ,difficult breathing
- ✓ Air hunger and restlessness
- ✓ Stridor
- ✓ DEATH unless proper airway management
- ✓ If suspected epiglottitis do not examine the throat, immediately take the patient to the O.R and intubate the patient

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Otitis Media



Cont'd

- **Otitis Media** is an inflammatory condition of the middle ear that results from *dysfunction of the eustachian tube* in association with a number of illnesses, including URIs and chronic rhinosinusitis
- The inflammatory response to these conditions leads to the development of *a sterile transudate* within *the middle ear* and *mastoid cavities*.
- *Infection may occur if bacteria or viruses from the nasopharynx contaminate this fluid*

Cont'd

- Acute OM
 < 3 weeks
- - Subacute OM
 3 weeks to 3 months
- - Chronic OM
 3 months or longer

Acute Otitis Media

- Is Acute infection of the mucous membrane lining of the middle ear
- It results when pathogens from the nasopharynx are introduced into the inflammatory fluid collected in the middle ear
- Most common in infants 6 – 18 months old (2/3 of cases)
- **Route of infection:** - Eustachian tube
 - External auditory canal
 - Blood borne

Etiology

Bacteria:-

- **S. pneumoniae - 30-35%**
- H. influenzae - 20-25%
- M. catarrhalis - 10-15%
- Group A strep - 2-4%

Viruses:-

- Respiratory syncytial virus (RSV)
- Rhinovirus
- Parainfluenza virus
- Influenza virus

Stages of AOM

1-tubal occlusion



2-pre-suppurative



3-suppurative



5-complication



4-resolution

1-tubal occlusion

Edema and hyperemia of nasopharyngeal end of the eustachian tube → block the tube → absorption of air and -ve pressure → retraction and some degree of effusion

Symptoms

- 1- mild earach
- 2- mild deafness
- 3- no fever

Sign:

- 1- tympanic membrane retracted
- 2- loss of light reflex
- 3- tuning fork show (CHL)

2-pre-suppurative

If tubal occlusion is prolonged → pyogenic organism invade

Symptoms:

- 1-earache
- 2-deafness
- 3-tinnitus
- 4- fever

Sign:

- 1-congestion and redness of the tympanic membrane
- 2-tuning fork (CHL)

3-suppuration

- 1-formation of pus
- 2-bulging of tympanic membrane

Symptoms

- 1- very severe earache
- 2-marked deafness
- 3-very high fever+vomiting

Sign

Redness
+bulging of
tympanic
membrane+loss
of land mark

L

4-Resolution

- Rapture of TM
- release of pus
- subsidence of symptoms

Note:

If the treatment is started early → resolution may start without rapture of TM

Symptoms

- 1- earach is relived
- 2- pus discharge from the ear
- 3- fever comes down (child feel better)

Sign

- 1- small perforation in antero-infero of para tensa
- 2- return normal color and land mark of TM

Complications of otitis media

➤ **Intracranial:**

- ✓ **Meningitis**
- ✓ Epidural abscess
- ✓ Brain abscess
- ✓ Cavernous sinus thrombosis
- ✓ Lateral sinus thrombosis
- ✓ Subdural empyema
- ✓ Carotid artery thrombosis

➤ **Intratemporal :**

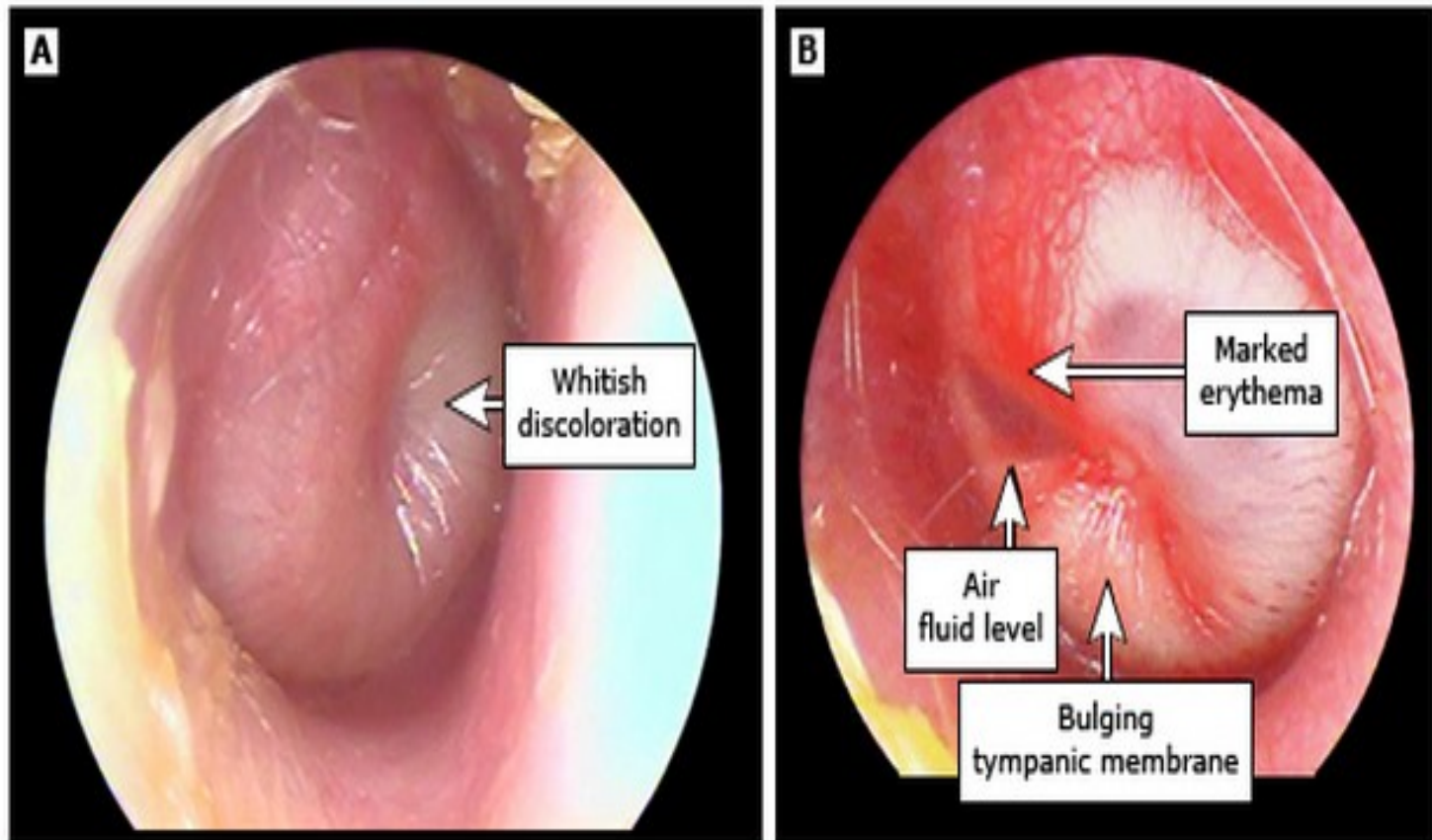
- Hearing loss
- Balance and motor problems.
- TM perforation.
- Cholesteatoma
- Adhesive otitis media
- Extension of the suppurative process to adjacent structures (mastoiditis, petrositis, labyrinthitis)

Sign and symptoms

- Ootalgia
- Fever
- deafness
- otorrhea



Acute otitis media



Chronic otitis media

- **Chronic Suppurative Otitis Media** is characterized by *persistent* or *recurrent purulent otorrhea in the setting of tympanic membrane perforation*
- Usually, there is also some degree of **conductive hearing loss.**

Acute laryngitis

- *It is inflammation of laryngitis*
- *Acute laryngitis can result from*
 - ✓ *Inhalation of irritants*
 - ✓ *Viral infections*
 - ✓ *Bacterial infections*
- **Two forms of laryngitis:**
 - 1. Tuberculous**
 - 2. Diphtheritic**

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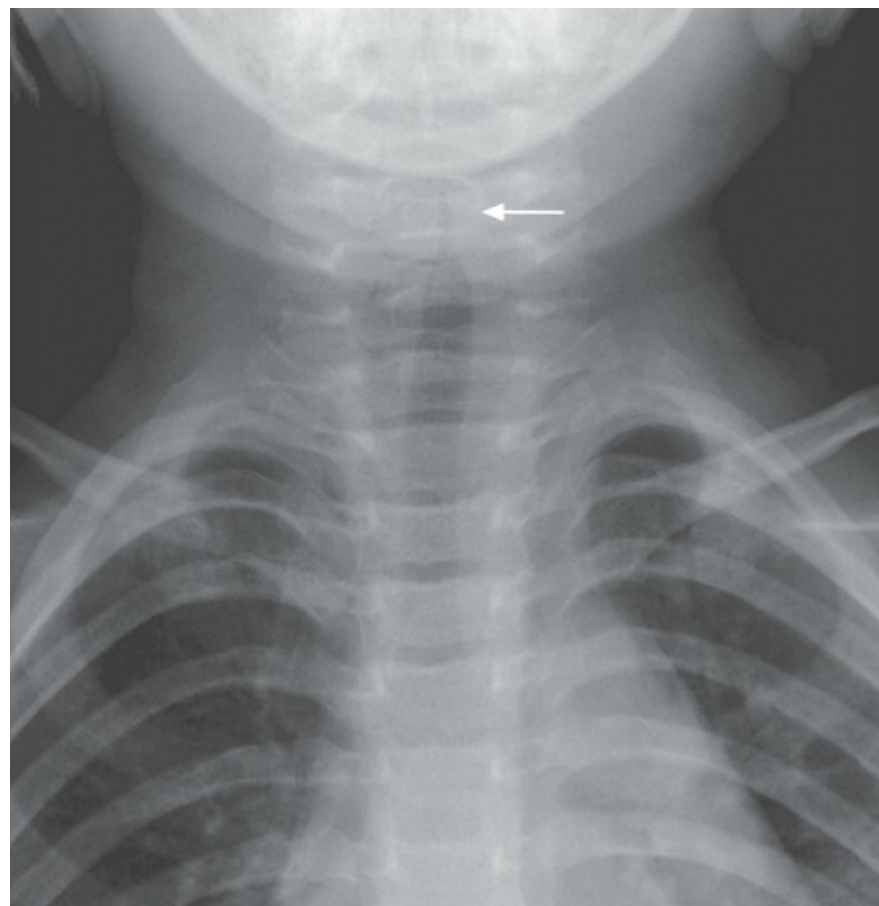
- ***Tuberculos laryngitis***: consequence of protracted active tuberculosis, during which infected sputum is coughed up.
- ***Diphtheritic laryngitis***: caused by *Corynebacterium diphtheriae* implants on the mucosa of the upper airways
- Exotoxin causes necrosis of the mucosal epithelium and a dense fibrinopurulent exudate (dirty-gray pseudomembrane of diphtheria)
- Aspiration of the pseudomembrane causing obstruction of major airways and absorption of bacterial exotoxins

CROUP (laryngotracheobronchitis)

- Etiology , most are viral :
 - ✓ Parainfluenza(1,2,3)-75%
 - ✓ Others ,Influenza A and B) ,adenovirus,RSV ,measles .
 - ✓ Mycoplasma pneumoniae (rare)
- Age : between 5months and 5 years, peak in 2nd year of life.
- Males > females. More in winter

Clinical manifestations

- Rinorrhea
- Pharyngitis
- Low grade fever for 1-3
- Barking cough
- Hoarseness
- Inspiratory stridor



Nasopharyngeal carcinoma

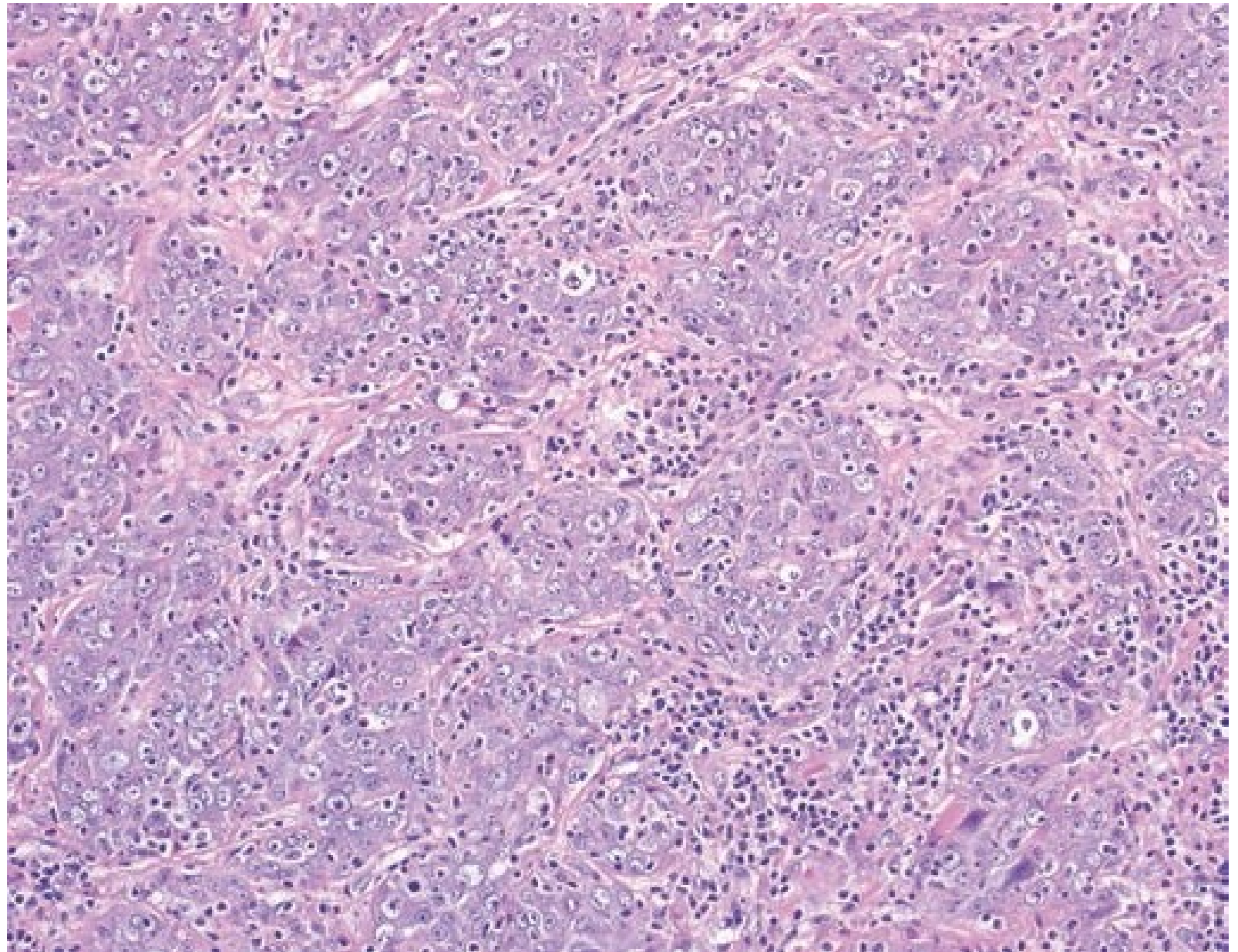
- Nasopharyngeal carcinoma (NPC) is a carcinoma arising in the nasopharyngeal mucosa that shows histologic or immunophenotypic of squamous differentiation
- The strong epidemiologic links to **EBV** and the high frequency of this cancer among the Chinese
- Mostly it spread to cervical lymph nodes, and

classification of nasopharyngeal carcinoma

1. Nonkeratinizing squamous cell carcinoma
 - *Differentiated subtype*
 - *Undifferentiated subtype*
2. Keratinizing squamous cell carcinoma
3. Basaloid squamous cell carcinoma

Cont'd

- Undifferentiated is the most common subtype, accounting for > 60% of nasopharyngeal carcinoma
- Peak incidence in fourth to sixth decades; less than 20% occur in pediatric age group
- The undifferentiated neoplasm is characterized by large epithelial cells with indistinct cell borders (reflecting “**syncytial**” growth) and **prominent eosinophilic nucleoli**.
- Accompanied by a striking **influx of T cells**



Presenting features	Frequency
Symptoms	
Neck mass	42%
Nasal (postnasal drip, discharge, bleeding, obstruction)	46%
Aural (tinnitus, discharge, earache, deafness)	42%
Headache	16%
Double vision, squint, blindness	6%
Facial numbness	5%
Speech/swallowing problem	2%
Weight loss	4%
Physical signs	
Enlarged neck node(s)	72%
Bilateral neck nodes	35%
Neck nodes extending to supraclavicular fossa	12%
Cranial nerve palsy	10%
Deafness	3%

Laryngeal Tumors

- variety of non neoplastic, benign, and malignant neoplasms of epithelial and mesenchymal origin may arise in the larynx
- vocal cord nodules, papillomas, and squamous cell carcinomas are common
- The most common presenting feature is hoarseness.

Nonmalignant Lesions

1. Vocal cord nodules (“polyps”): smooth, hemispherical protrusions located on the **true vocal cords**.

- Composed of fibrous tissue and covered by stratified squamous mucosa
- These lesions occur chiefly in heavy **smokers** or **singers**

2. Laryngeal papilloma or squamous papilloma

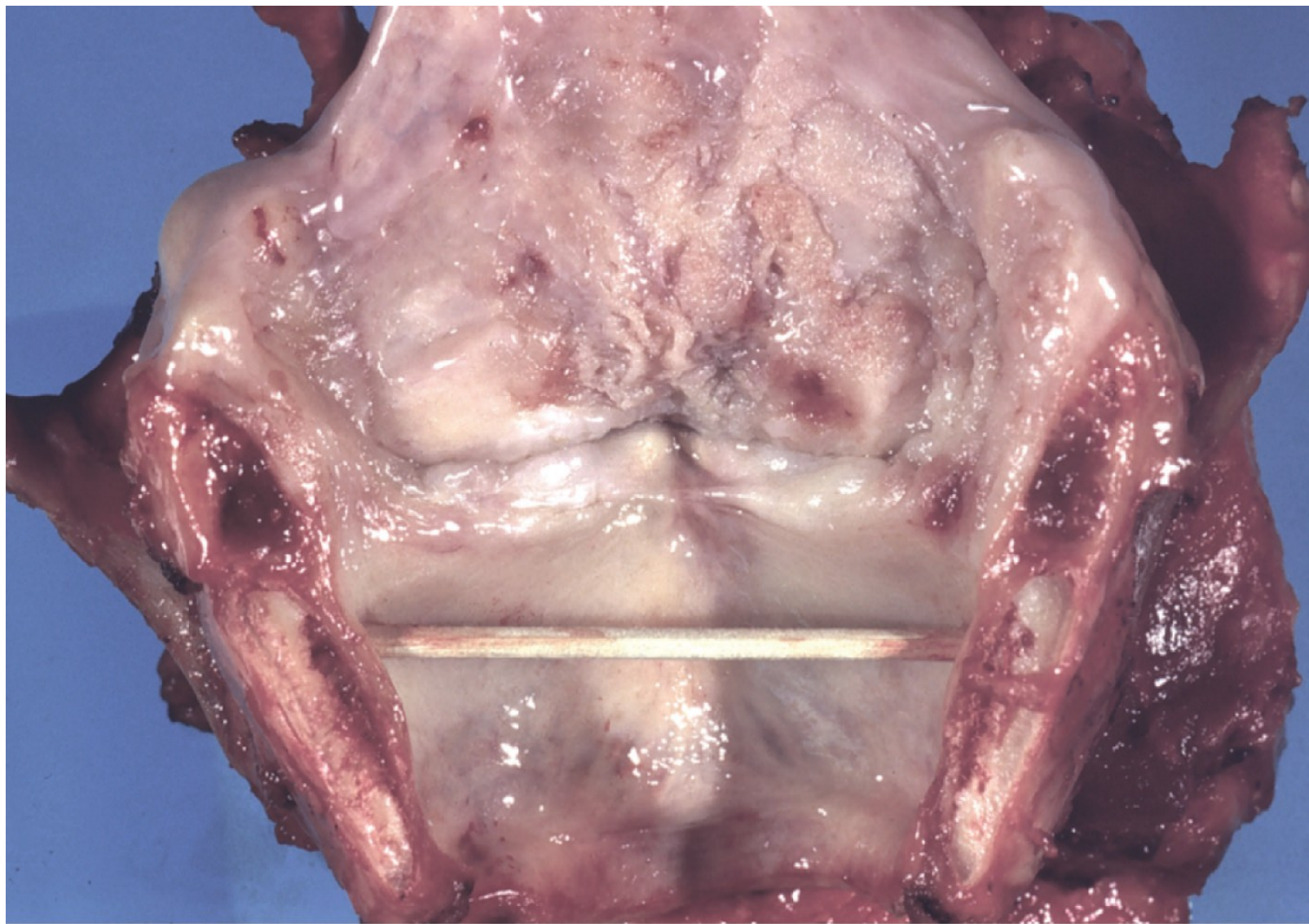
- Soft raspberry like excrescence benign neoplasm usually located on the true vocal cords
- They are single in adults but often are multiple in children(as *recurrent respiratory papillomatosis*)
- These lesions are caused HPV types 6 and 11 and often spontaneously regress at puberty.
- Cancerous transformation is rare.

Carcinoma of the Larynx

- Carcinoma of the larynx represents only 2% of all cancers
- most commonly occurs after 40 years of age and is more common in men than in women
- Risk factors: smoking
 - Alcohol
 - Asbestos exposure
 - HPV

Cont'd

- 95% of laryngeal cancers are squamous cell carcinomas
- Tumor location:
 - ✓ vocal cords (glottic tumors) in 60% to 75%
 - ✓ supraglottic; 25% to 40%
 - ✓ subglottic; <5%
- The location of the tumor within the larynx has a significant prognostic effect



Thank You